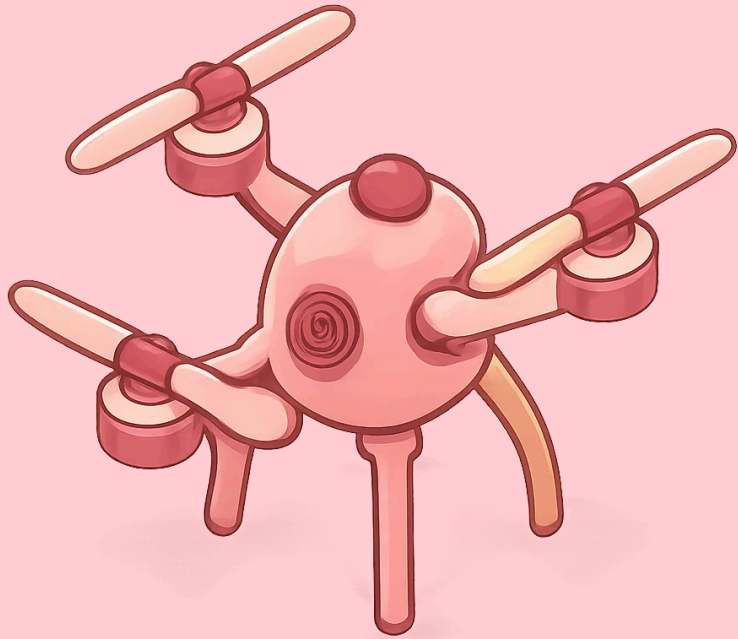




PROJECT REVIEW – I

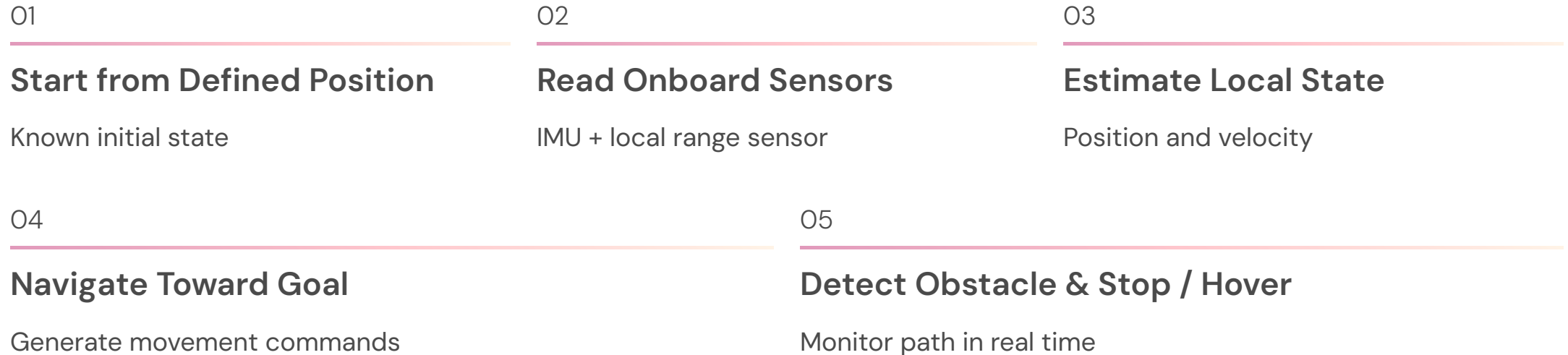
ROS 2 · AUTONOMOUS NAVIGATION

Autonomous Drone Navigation

A ROS 2-based MVP for GPS-independent local navigation — demonstrating sensor fusion, state estimation, waypoint navigation, and obstacle response within a course project scope.

Core MVP Concept

A drone navigating **without GPS** — from start to goal, detecting and responding to obstacles using onboard sensing and ROS 2 coordination.



Why GPS-Independent Navigation?

GPS is unavailable or unreliable in many real-world environments. Onboard sensing provides a path to autonomous local navigation.

→ GPS is Not Always Available

Indoors, urban canyons, and underground environments block or degrade GPS signals.

→ IMU Provides Continuous Motion Data

Accelerometers and gyros deliver high-frequency motion information — but drift accumulates over time.

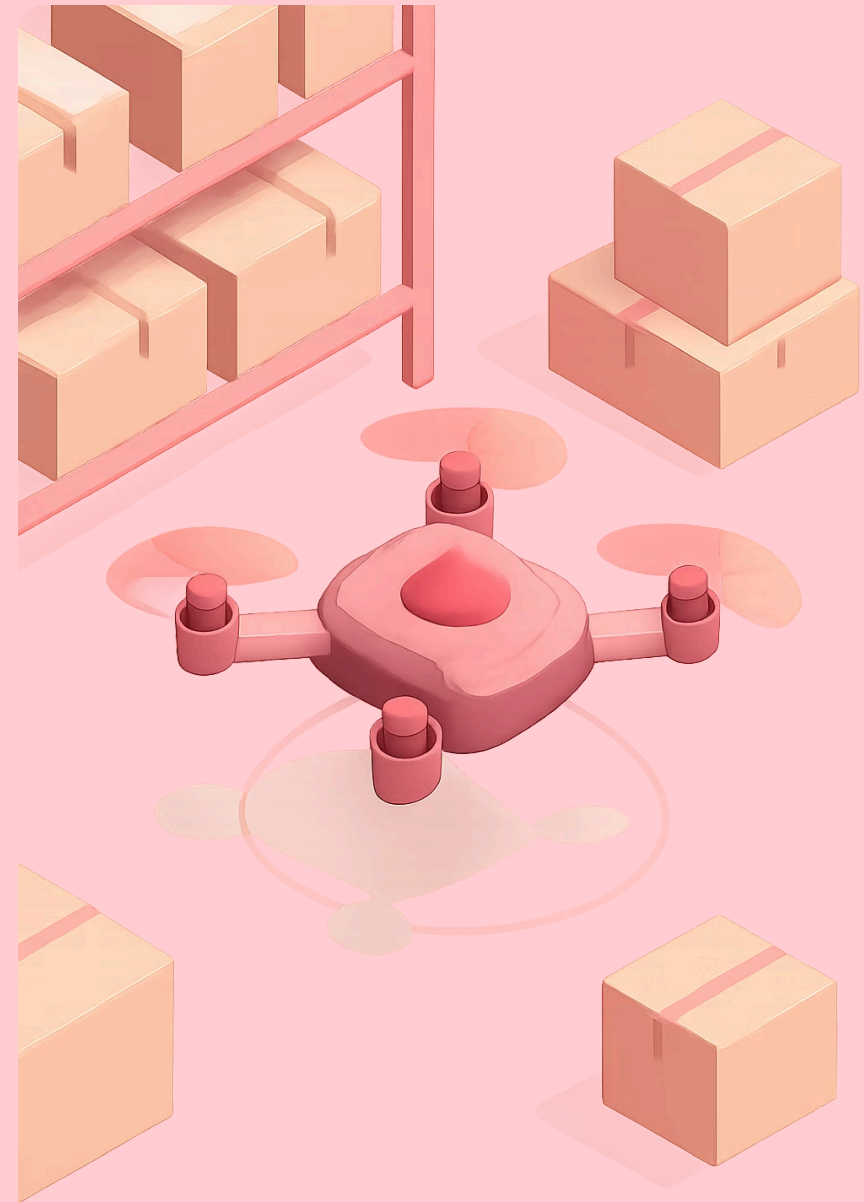
→ Complementary Sensing Improves Estimation

A local range or optical-flow sensor constrains drift and improves local state estimates.

→ ROS 2 Enables Modular Integration

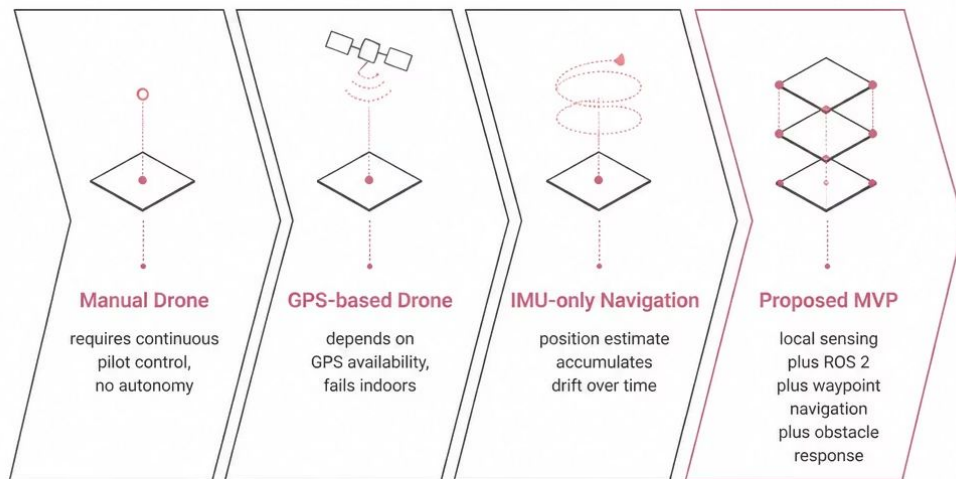
Sensing, estimation, navigation, and control connect cleanly through topics and services.

❑ **MVP Goal:** Demonstrate a drone moving from START → DESTINATION while reacting to a simple obstacle — without using GPS as the navigation source.



Problem Statement

"How can a small autonomous drone navigate from a defined start position toward a destination using onboard sensor data, without relying on GPS, while responding safely to a simple obstacle?"



Where the MVP Fits

Each approach has a clear limitation. The MVP bridges the gap — using local sensing and ROS 2 to achieve basic autonomous navigation without GPS dependency.

- **Manual:** No autonomy
- **GPS-based:** Environment-dependent
- **IMU-only:** Drift accumulates
- **MVP:** Local sensing + ROS 2 + obstacle response

What Existing Research Tells Us

Chang et al. — 2023

GPS-denied UAV review — navigation requires sensing, localization, and planning.

Youn et al. — 2021

Low-cost GPS-denied UAV — sensor fusion supports autonomous indoor flight.

Lin & Zhan — 2022

Visual-inertial + UWB — complementary measurements constrain localization drift.

Key Literature Insight

IMU Alone → Drift

Position error grows without correction

Complementary Sensing → Better Estimation

Range or optical flow constrains drift

Obstacle Sensing → Safety

Reactive response prevents collisions

ROS-based Architecture → Integration

Modular nodes connect the pipeline

What Are We Actually Building?

The Engineering Gap

The individual technologies are established. The MVP focuses on **integrating them into a simple, demonstrable ROS 2 navigation pipeline** within a course project.

MVP Objectives

- 1 **Modular ROS 2 Node Architecture**
- 2 **Sensor Fusion & State Estimation**
- 3 **Waypoint Navigation**
- 4 **Obstacle Detection & Response**
- 5 **Demonstrate ROS 2 Communication**

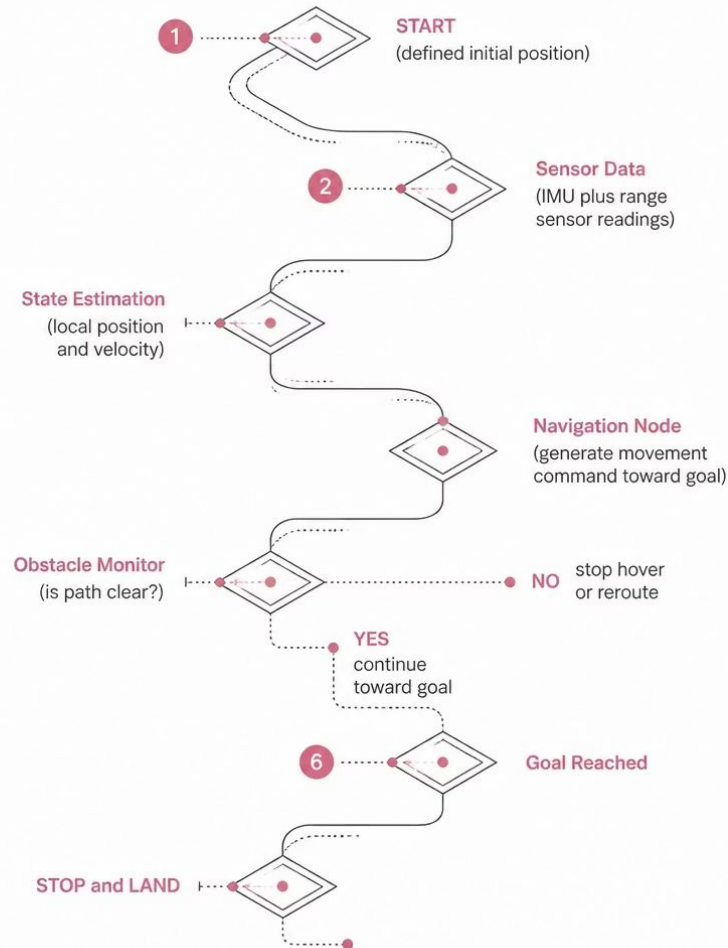
Research-Grade Systems

Complex sensors + SLAM + advanced planning

Our MVP

IMU + local sensing + ROS 2 + waypoint nav + obstacle response

Autonomous Navigation Workflow



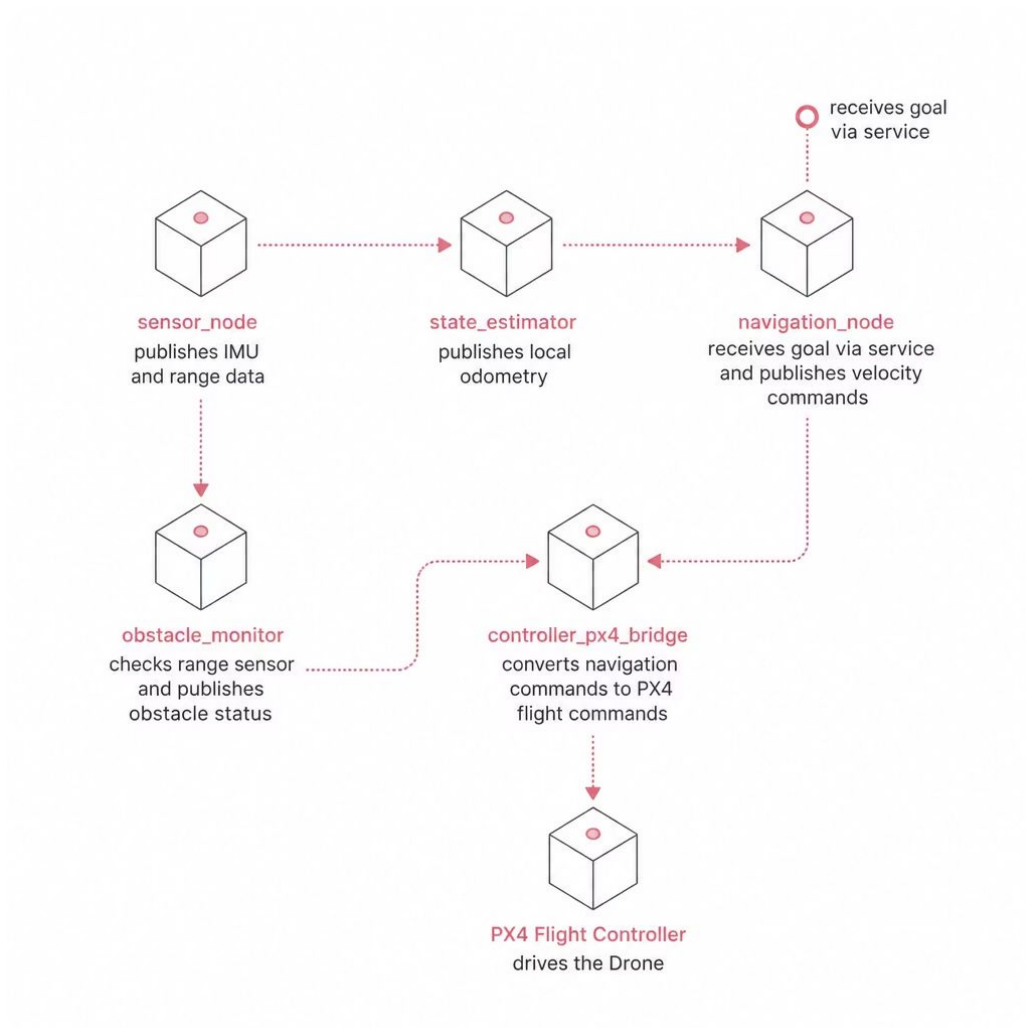
A Simple, Closed-Loop Pipeline

This is an intentionally scoped MVP navigation stack — not a production-grade autonomy system. The feedback loop runs continuously:

Each stage publishes or subscribes via ROS 2 topics, keeping modules decoupled and testable.

Sensors → ROS 2 → Navigation → Flight Command → Drone Movement → Sensors

ROS 2 Node Architecture



Node Responsibilities

ROS 2 Interfaces

Topics: /sensors/imu, /sensors/range, /state/odom, /navigation/cmd, /navigation/obstacle

Service: /navigation/set_goal

Action (optional): /navigation/navigate_to

sensor_node

Publishes IMU and range data

state_estimator

Estimates local position and velocity

navigation_node

Generates movement commands toward goal

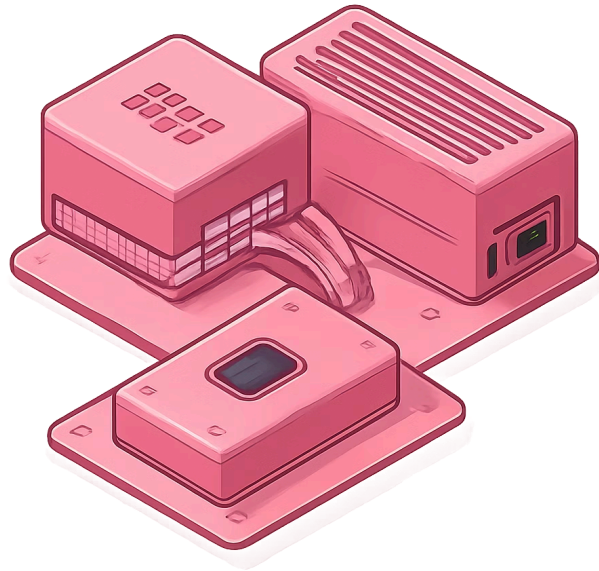
obstacle_monitor

Checks path and signals blockage

controller / px4_bridge

Converts commands to flight instructions

Minimal Hardware Architecture



Core Hardware Stack

Quadrotor Platform

Standard multirotor frame

PX4 Flight Controller

Handles low-level stabilization

Companion Computer

Runs ROS 2 nodes

Sensors

IMU + ToF / LiDAR + optional optical flow

- ❑ **Scope Statement:** The MVP prioritizes demonstrating the ROS navigation architecture. Hardware complexity is intentionally minimized — this is not a component procurement review.

Expected Outcome & Conclusion

The MVP demonstrates a **functional ROS 2-based autonomous navigation pipeline** — scoped for a course project, grounded in established research, and focused on integration over complexity.

Modular ROS 2 Architecture

Decoupled nodes communicating via topics and services

IMU + Local Sensor Fusion

Onboard sensing replaces GPS for local navigation

Simple Obstacle Response

Stop, hover, or reroute when path is blocked

PX4-Compatible Interface

Flight controller bridge for real-world demonstration

This is a **Minimum Viable Prototype** — intentionally scoped to demonstrate core ROS concepts within a course timeline, not a production-ready autonomous system.

